MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
FIRST SEMESTER EXAMINATION
(2024 ADMISION ONWARDS)

MG1DSCZIM102 – Fundamentals of Biochemistry

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer **All** Questions

Each question carries **1** mark

1.Theory of spotaneous generation shows life originates from…… [Remember] [CO1]
2.……..is the negative logarithm of H+ ions. [Understand] [C02]
3.Double helical model of DNA was proposed in the year…… [Remember] [CO1]
4.The four hydrogen bond in water molecules are arranged tetrahedrally to form…..structure. [Apply][CO2]
5.The prominent role of mitochondria is to produce energy in the form of.....… [Understand] [C04]
6 .....proposed fluid Mosaic model in 1972 [Remember] [CO3]
7.Prokaryotes have .....ribosomes ,each consist of 30 S and 50 S subunits. [Analyse] [C03]
8.Lysosomes are frequently nicknamed as.....… [Remember] [CO4]
9.During symbiotic nitrogen fixation cellwall of roothair invaginates to form..... [Evaluate] [CO5]
10 .Monoterpenes has…….isoprene units. [Remember] [CO5]

[1 x 10 = 10]

Part B

Short Questions

Answer10 **Q**uestions

Each question carries 2 marks

11. Differentiate between biogenesis and abiogenesis. [Remember] [CO1]
12. Outline the contribution of Louis Pasteur. [Remember] [CO1]
13. Write short note on major forces stabilizing water molecule. [Apply] [CO2]
14. Explain the difference between strong and weak acid. [Understand] [CO2]
15. Describe the key feature of fluid mosaic model of plasma membrane? [Understand] [CO3]
16. Explain Characteristics of peripheral membrane proteins with one example. [Apply] [CO3]
17. Define Active transport . [Remember] [CO4]
18. Write a note on Rough Endoplasmic Reticulum. [Apply] [CO4]
19. Write short note on thylakoids [Apply] [CO5]
20. Describe Cyclic photophosphorylation [Remember] [CO5]
21. Discuss the morphology of Endoplasmic Reticulum. [Understand] [CO4]
22. Explain Root nodulation process in nitrogen fixation. [Understand] [CO5]

[2 x 10 = 20]

Part **C**

Short Essay Type Questions

Answer 5 Questions

Each question carries 4 marks

23. Draw structure of water molecule with their bond angle. [Remember] [CO2]
24. Explain the significance of keymilestones in the history of biochemistry. [Remember] [CO1]
25. Explain the concept of pH Scale,its importance and application in various fields. [Apply] [CO2]
26. Explain fluid mosaic model of plasma membrane. [Understand] [CO3]
27. Compare and contrast isotonic,hypotonic and hypertonic solution.[Apply] [CO4]
28. Discuss the significance of Mitosis & Meiosis. [Understand] [CO4]
29. Write an essay on molecular mechanism involved in symbiotic nitrogen fixation [Analyse] [CO5]

[4 x 5 = 20]